

Collective Logic Lab - Setting up Honey Bee Project

Notes:

- Documentation by Diya Hamada (dhamada@asu.edu) on M2 Mac.
- Updated Dec 12th, 2025

Step 1: Git clone the honey-bee-behavior repository.

Link: <https://github.com/Collective-Logic-Lab/honey-bee-behavior/>

Ask Dr. Daniels for help if you run into any access issues.

Go to your terminal, "cd" into the folder you would like this repo to live in, and run

```
"git clone https://github.com/Collective-Logic-Lab/honey-bee-behavior.git"
```

Step 2: Download relevant data and files to your computer.

For the Honey Bee Project, the data files are in Zenodo:

2018 data (more commonly used): <https://zenodo.org/records/6045860>

2019 data: <https://zenodo.org/record/7298798>

[beetrajectories_days_000_to_049.zip](#), [beetrajectories_days_050_to_085.zip](#)

Single-day hdf files, each containing x-y trajectory data in the form:

daynum	framenum	uid	x	y	camera	theta
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- daynum: the day number of the observation period. See 'Day numbers and cohorts' sheet in 2018_Quantity_Descriptions.xlsx
- framenum: using the camera frame rate of 3fps, the frame number with respect to that particular day
- uid: the Unique ID of each bee
- x,y: pixel coordinate values of each bee. Conversion is 80 pixels/cm
- camera: which camera the bee was detected on: 0=exit side (with dance floor - shown at right in comb map images), 1=back side (shown at left in comb map images)
- theta: orientation of the bee in the hive

Age	Day number	Bee unique ID	Cohort ID	<metric1>	<metric2>	...
df_day1min_alldays.zip , df_day5min_alldays.zip , df_dayhour_alldays.zip						
Behavioral metrics calculated over different time intervals: 1 minute, 5 minute, or 1 hour divisions. See 2018_Quantity_Descriptions.xlsx for full list and detail of quantities. Files are organized as a single .hdf file for each day. Each file contains data in the form:						

Just click the .zip folder(s) you want. If you're unsure, the 2018 data with days 0-49 (beetrajectories_days_000_to_049.zip) and the 1-minute interval df_day data (df_day1min_alldays.zip) are a good starting point. Downloading and unzipping takes a while. Move the unzipped folders into the cloned honey-bee-behavior folder.

Step 2: Make your own branch

Always check that you're starting from the latest version of the main branch! In your terminal, having "cd"ed into honey-bee-behavior, type "**git checkout main**" and "**git pull origin main**".

Now create your branch and switch to it. "**git checkout -b <branch name>**"

"**git branch**" should now show your branch and have a little star next to it!

Step 3: Open Jupyter Lab

We have two methods for opening Jupyter Notebooks: Dr. Daniels is keen on Anaconda, while I use my Terminal.

Anaconda:

Download the Anaconda Navigator:

<https://www.anaconda.com/products/navigator>

You can simply click on JupyterLab from the Anaconda Navigator interface and navigate to the honey-bee-behavior folder using the folder navigator on the left of the JupyterLab interface.

Terminal:

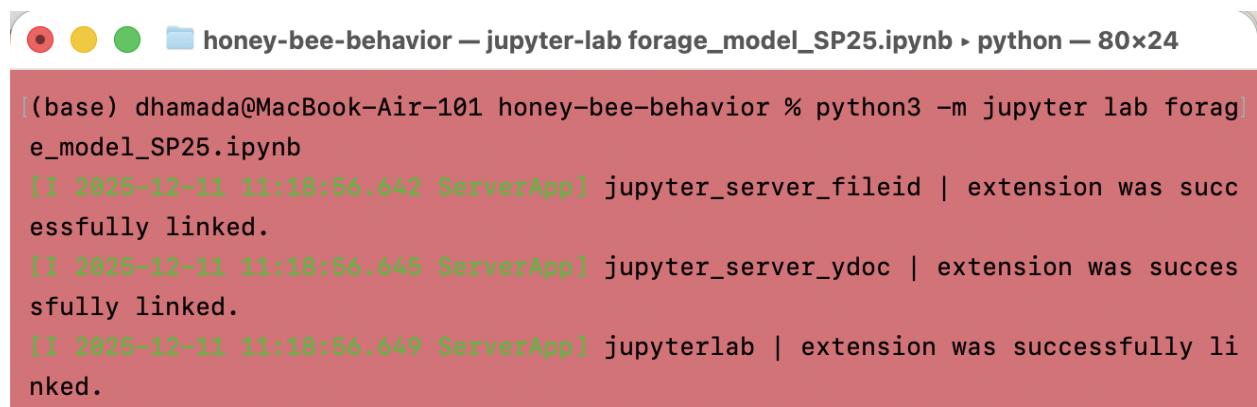
You will need Python on your computer—I use Python3. If you need help with that, I recommend checking this link out:

<https://realpython.com/installing-python/>

You may want JupyterLab. Here's another helpful link:

https://jupyterlab.readthedocs.io/en/stable/getting_started/installation.html

Open your Terminal and navigate to the honey-bee-behavior folder using "cd". Then type **"python3 -m jupyter lab [ipynb file name]"** as shown below.

A screenshot of a macOS Terminal window. The title bar shows three colored window control buttons (red, yellow, green) followed by the text "honey-bee-behavior — jupyter-lab forage_model_SP25.ipynb › python — 80x24". The terminal content shows a user prompt "(base) dhamada@MacBook-Air-101 honey-bee-behavior %" followed by the command "python3 -m jupyter lab forage_model_SP25.ipynb". The output consists of three lines of green text: "[I 2025-12-11 11:18:56.642 ServerApp] jupyter_server_fileid | extension was successfully linked.", "[I 2025-12-11 11:18:56.645 ServerApp] jupyter_server_ydoc | extension was successfully linked.", and "[I 2025-12-11 11:18:56.649 ServerApp] jupyterlab | extension was successfully linked.".

```
(base) dhamada@MacBook-Air-101 honey-bee-behavior % python3 -m jupyter lab forage_model_SP25.ipynb
[I 2025-12-11 11:18:56.642 ServerApp] jupyter_server_fileid | extension was successfully linked.
[I 2025-12-11 11:18:56.645 ServerApp] jupyter_server_ydoc | extension was successfully linked.
[I 2025-12-11 11:18:56.649 ServerApp] jupyterlab | extension was successfully linked.
```

A ton of text will run across your Terminal, and then the notebook will open in JupyterLab in your browser.

"python3 -m jupyter lab" also works fine by itself, you'll just need to go to the folder navigator on the left of JupyterLab and select the file you want to open or make a new one :)

Step 4: Push your changes to your branch (and possibly make a pull request!).

Make sure all your changes are saved and head back to your Terminal (use "cd" to get back into honey-bee-behavior if you aren't already there).

Then you can decide what changes you'd like to see made in the original repository - type "**git status**" to see the modifications you've made. Be sure to unstage the data folders or files you added—there's just not space to fit them in the repo, so keep them local.

"**git restore --staged <file>**" to unstage a file

"**git add <file>**" to stage a file

Once you're happy with the set of staged files, write your commit message and push to your branch.

"**git commit -m "little note about what changes you made"**"

"**git push -u origin <branch name>**"

Are you at a stage in which you would like to merge with the main branch? Make a pull request. Pop over to GitHub, go to your branch, and hit "Compare & pull request". Be sure to be super descriptive in your message for clear documentation :)

Notes for JupyterLab:

- *Save frequently. Just in case. command+S or File>Save Notebook.*
- *Don't be afraid to restart the kernel when you are debugging or updating other files your notebook is dependent on (Kernel>Restart Kernel).*